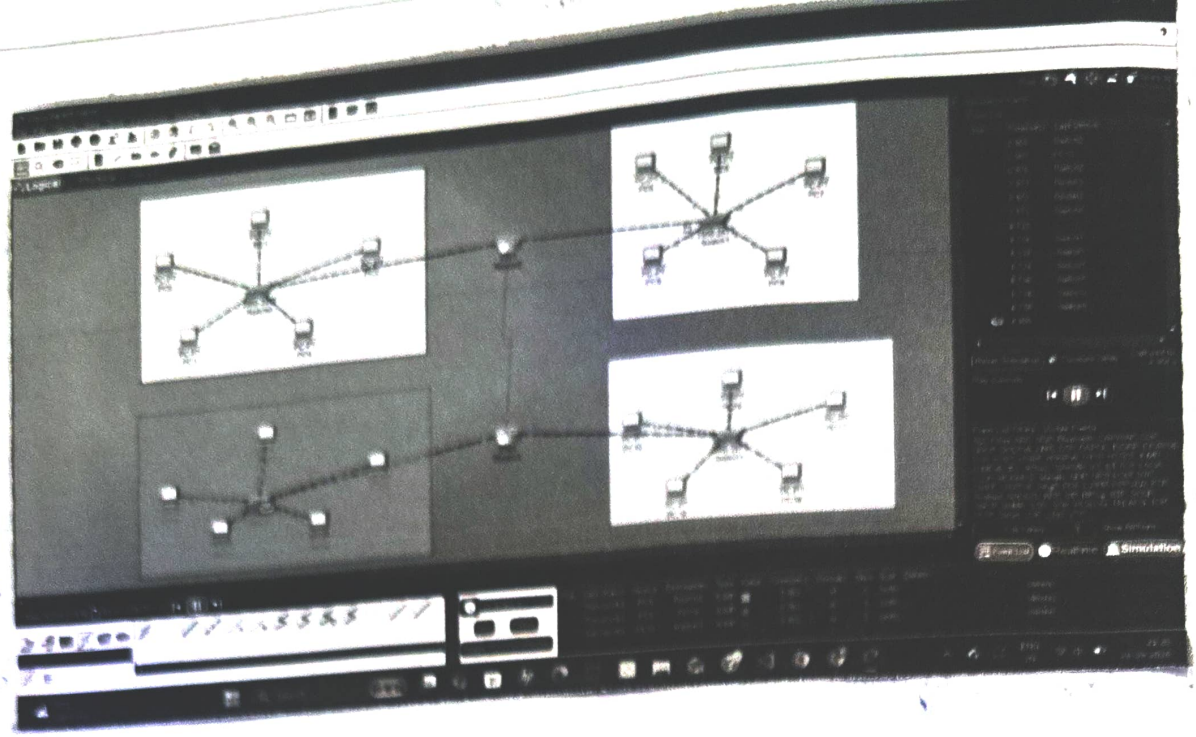
Experiment - 9

Aim:
Implementation of subnetting in Cisco Packet Tracer simulator.

classless IP subnetting is a technique that allows for more efficient use of IP address by allowing for subnet masks that are not just the default masks for each IP class.

OBSERVATION:

1) A network topology is created in Cisco Packet Tracer using routers, switches and PCs
2) The network address 192.168.1.0/24 is divided into smaller subnets.
3) A/ 127 subnet mask is used for subnetting.
4) The /27 mask provides 8 subnets, with 30 usable Host addresses in each subnet
5) Suitable IP address subnet mask and default gateways are assigned
6) Router interfaces are configured using the appropriate IP addresses and subnet masks
7) The switch ports are in access mode.
8) successful ping replies indicate that the subnetted network is function correctly.



Example IP address:

Router: 192.168.1.1

PC1 : 192.168.1.11

PC2 : 192.168.1.12

PC3 : 192.168.1.13

Subnet mask : 255.255.255.224 (/27)

Result

Thus, classless IP subnetting was successfully implemented in Cisco packet tracer.